

16/09/25

		a	0
	a	b	1
	a	b	8 1
a	b	8	$1 + 1 = 2$
a	b	8	$1 + 2 = 3$
	b	8	$2 + 3 = 5$
		8	$3 + 5 = 8$

Swapping of two numbers

a = 5

b = 7

Print(a, b)

a, b = b, a

Print(a, b)

o/p :-

5 7

7 5

Write a program for Fibonacci Series of 1st 10 values
Using while loop.

Note:- Fibonacci Series always start with 0 and 1

a, b = 0, 1

i = 1

while (i <= 10):

Print (a)

s = a + b

a = b

b = s

i = i + 1

o/p:- 0

1

1

2

3

5

8

13

21

34

o/p

0

1

1

2

3

...

...

...

...

34

Logic

Print (0)

s = a + b

a = b

b = s

i = i + 1

0 - first a value

s = 0 + 1 = 1

a = b - for a we are storing
b so present b
value is 1, then
a become 1

b = s

b = 1 - present s value.

loop repeat 8

Print a value 1

1

s = 1 + 1 = 2

a = b

a = 1

b = 2

again loop repeat

1

1 + 2 = 3

a = 2

b = 3

again loop repeat

a = 2

2 + 3 = 5

a = 3

b = 5

again loop repeat

Note:- Even numbers start from 2
odd numbers start from 1, 3
Fibonacci numbers start from 0, 1

Write a Program to calculate sum of digits without function & using while loop

Sample Output

257
14 i.e. $2+5+7$

num = int(input())

sum = 0

while (num > 0):

digit = num % 10

sum = sum + digit

num = num // 10

Print(sum)

o/p:-

257

14

logic

Think math logic

In 257 I want 7

$257 \% 10$

257 % 10

7

next I should get 25

$257 // 10$

25

Again I do

$25 \% 10$

5

Again $25 // 10$

2

initial num value is 257

num = 257

initial sum is 0

sum = 0

condition is num > 0 when num < 0 it comes out from loop

$257 > 0$ - com. inside loop

$257 \% 10 = 7$

$0 + 7 = 7$

$257 // 10 = 25$ - this 25 again stores back to num

loop continue

num = 25

sum = 7

$25 \% 10 = 5$

$$4 + 8 = 12$$

$$25 // 10 = 2$$

again this 2 stored to num

$$\text{num} = 2$$

$$\text{sum} = 12$$

$$2 > 0$$

$$2 \% 10 = 2$$

$$12 + 2 = 14$$

$$2 // 10 = 0$$

again this 0 stored to num

$$\text{num} = 0$$

$$\text{sum} = 14$$

$0 > 0$ - False. it will stop executing

finally I am printing final sum value.

Note: when ever any output is look like row & column format in the. Since we have to use nested loop
→ if it is only one row or one column, we can execute in single loop

Nested loop

1) for loop inside for loop

for i in range(): # outer loop

for j in range(): # inner loop

Statements of inner loop

Statements of outer loop

Syntax for
for loop
inside a
for loop

1	2	3	4	5	6	7	8	9	10
2	4	6	8	10	12	14	16	18	20
3	6	-	-	-	-	-	-	-	-
4	8	-	-	-	-	-	-	-	-
5	10	-	-	-	-	-	-	-	-
6	12	-	-	-	-	-	-	-	-
7	14	-	-	-	-	-	-	-	-
8	16	-	-	-	-	-	-	-	-
9	18	-	-	-	-	-	-	-	-
10	20	-	-	-	-	-	-	-	-

i want to
Print o/p
like this

→ first observe how many rows are there in o/p & then observe in each row how many elements are printing
10 rows are there again in each row 10 elements are printing

Note:- for outer loop usually we mention how many rows
for inner loop how many elements i want in a each row

Code:-
for i in range(1, 11, 1):
 for j in range(1, 11, 1):
 Print(i, j), end = " " ^{top space}
 Print()

meaning don't go for
next line for printing
~~another~~ next value
give one top space
in the same line

Note: for and while is actually same but in for loop
we write in single line and in while loop we
write that single line in multiple lines

when $i=1$
 $j=1, 2, 3, 4, 5, 6, 7, 8, 9, 10$

$i=2$
 $j=1, 2, 3, 4, 5, 6, \dots, 10$

$i=3$

if multiply $i=1$ with all the value of j i will get
first line of o/p
1 2 3 4 5 6 7 8 9 10

★ For printing o/p like this

1	2	3	4	5
2				
3				
4				
5				
6				
7				
8				
9				
10				

for i in range(1, 11):

for j in range(1, 6):

Print($i * j$, end = " |t ")

Print()

Write a Program to print o/p like this

	2		4	
2	4	6	8	10
	6		12	
4	8	12	16	20
	10		20	

for i in range(1, 6):

for j in range(1, 6):

~~if $i * j$, end = " |t "~~

if $(i * j) \% 2 == 0$:

Print($i * j$, end = " |t ")

else:

Print(" ", end = " |t ")

Print()